Logarithmic Functions & Properties

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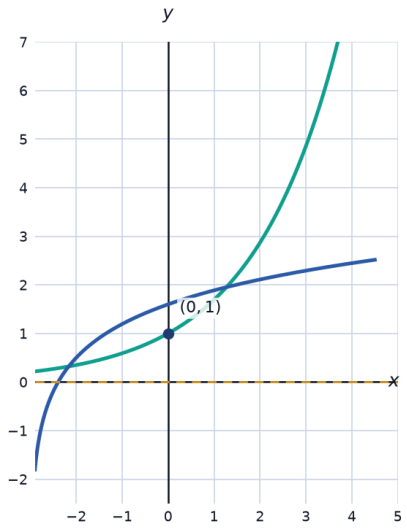
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting between exponential and logarithmic form**

Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

2 **[Foundational]** • **Target: Converting between exponential and logarithmic form**

Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.



3 **[Foundational]** • **Target: Properties of logarithms**

Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.

4 **[Foundational]** • **Target: Properties of logarithms**

Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

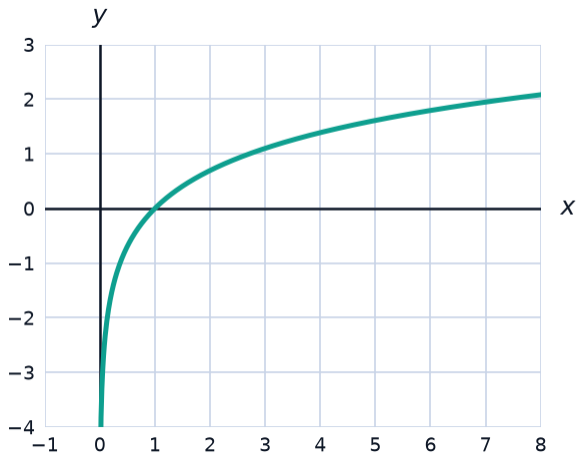
5 **[Foundational]** • **Target: Properties of logarithms**

Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

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6 [Foundational] • Target: Visualizing logarithmic functions

The graph shows $f(x) = \ln(x)$, the natural logarithm.



a State the domain and the vertical asymptote shown in the graph.

b State the x-intercept shown.

7 [Foundational] • Target: Solving exponential equations using logarithms

Solve $5^x = 40$ for x, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_2(x + 3) = 4$ for x, and check that the solution keeps the argument positive.

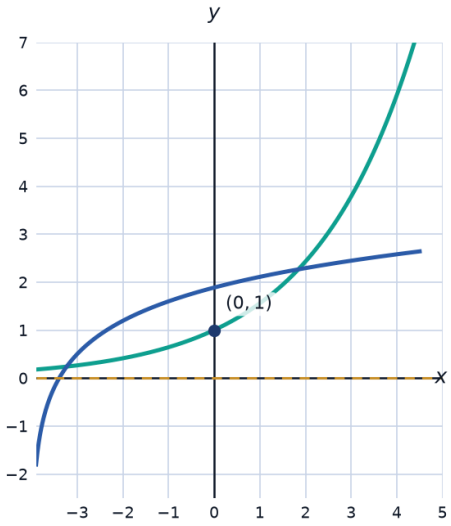
9 [Intermediate] • Target: Solving logarithmic equations

Solve $\log(x) + \log(x - 3) = 1$ for x, checking both solutions against the domain.

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10 **[Intermediate]** • **Target: Applications: the pH scale and decibels**

The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.

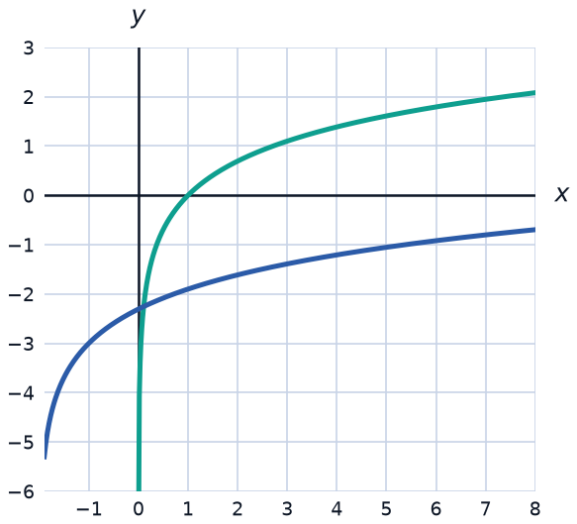


11 **[Intermediate]** • **Target: Applications: the pH scale and decibels**

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

12 **[Intermediate]** • **Target: Transformations of logarithmic functions**

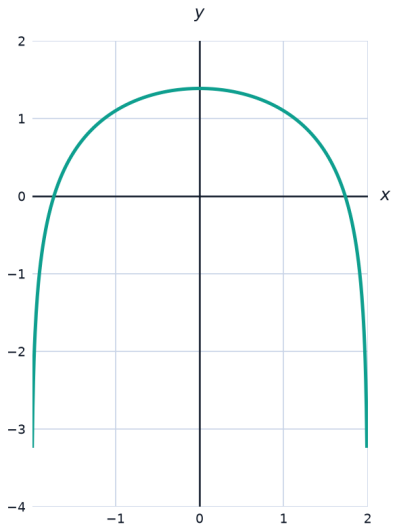
Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.



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13 [Intermediate] • Target: Transformations of logarithmic functions

Find the domain of $h(x) = \log(4 - x^2)$.



14 [Intermediate] • Target: Solving equations that combine exponentials and logarithms

Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

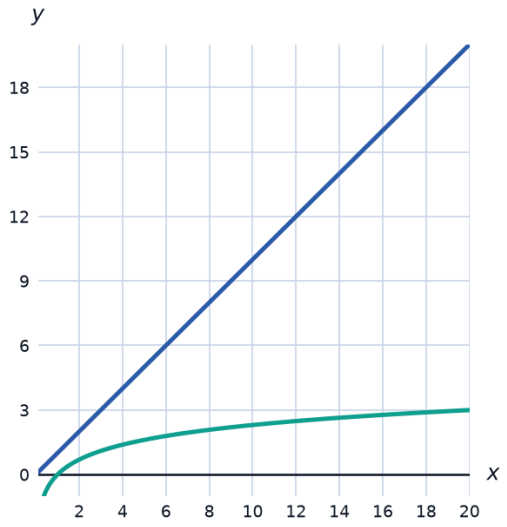
15 [Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve $\log_2(x) + \log_2(x - 2) = 3$ for x, checking the domain.

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16 [Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.



17 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 1.03 Rates of Change & Concavity

Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

18 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 2.01 Exponential Functions & Growth Models

If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.

19 [Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Level 1 · FoundationsOn Section I Part B of the AP Precalculus exam, a student allocates 50 minutes to answer 12 questions using a graphing calculator. What is the average time per question available in minutes, rounded to the nearest tenth?

20 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Converting between exponential and logarithmic form

Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10 times more intense than another, find the difference in their decibel levels.

a. 30

b. 20

c. 10

d. 1

22 [Foundational] • Target: Properties of logarithms

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

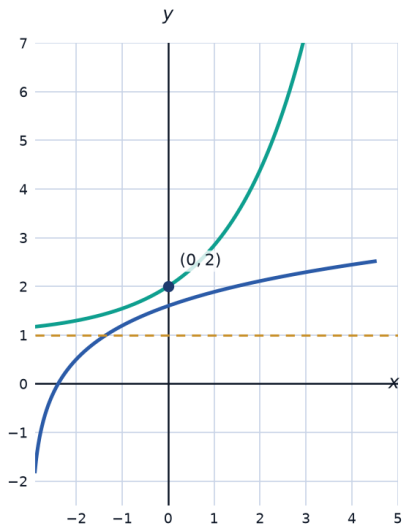
b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

23 [Foundational] • Target: Visualizing logarithmic functions

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.



a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

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24 [Foundational] • Target: Solving exponential equations using logarithms

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

25 [Foundational] • Target: Solving logarithmic equations

Solve: $2^{x+1} = 5$. Express x using logarithms.

a. $x = \frac{\ln 5}{\ln 2} - 1$

b. $x = \frac{\ln 5 - 1}{\ln 2}$

c. $x = \log_2 5 - 1$

d. Both A and C

26 [Foundational] • Target: Applications: the pH scale and decibels

Prove $a^{\log_a x} = x$ using natural logarithms.

a. Only for $a = e$

b. Not always true

c. Take $\ln$: $\ln(a^{\log_a x}) = \log_a x \cdot \ln a = \frac{\ln x}{\ln a} \cdot \ln a = \ln x$. So $a^{\log_a x} = e^{\ln x} = x$

d. By induction

27 [Foundational] • Target: Transformations of logarithmic functions

The function $f(x) = -2\cos(\frac{\pi}{3}x) + 5$ has amplitude, period, and midline

a. 2, 6, y=5

b. 2, 3, y=5

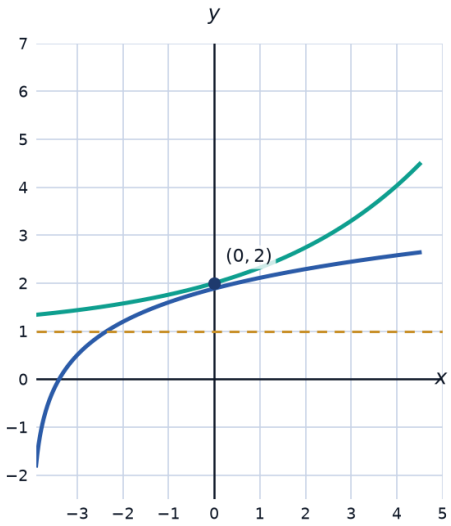
c. –2, 6, y=5

d. 2, 6, y=0

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28 [Foundational] • Target: Solving equations that combine exponentials and logarithms

A function is one-to-one if:



a. Every output repeats

b. Passes the vertical line test only

c. Passes the horizontal line test

d. Is a polynomial

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing logarithmic growth to other function families

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20

b. 24

c. 32

d. 44

30 [Intermediate] • Target: Modeling with logarithms: the Richter scale

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19

b. 31

c. 16

d. 11

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31 [Intermediate] • Target: Converting between exponential and logarithmic form

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$

b. $y = 0$

c. $y = 3$

d. $x = -1$

32 [Intermediate] • Target: Properties of logarithms

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. a = 4, b = 3

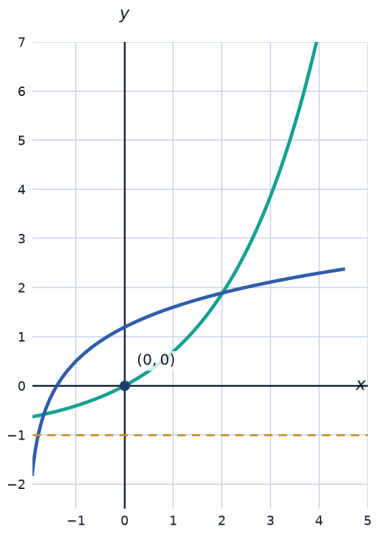
b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

33 [Intermediate] • Target: Visualizing logarithmic functions

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.



a. 2

b. 3

c. 4

d. 6

34 [Intermediate] • Target: Solving exponential equations using logarithms

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. 8

c. 9

d. 5

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35 [Intermediate] • Target: Solving logarithmic equations

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. 8

c. 6

d. 16

36 [Intermediate] • Target: Applications: the pH scale and decibels

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$

37 [Intermediate] • Target: Transformations of logarithmic functions

Convert to exponential form: $\log_5 25 = 2$.

a. $5^{25} = 2$

b. $2^5 = 25$

c. $5^2 = 25$

d. $25^2 = 5$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving equations that combine exponentials and logarithms • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$

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39 [Challenging] • Target: Comparing logarithmic growth to other function families • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Show that $e^{i\pi} + 1 = 0$ using Euler's formula.

a. $e^{i\pi} = i$

b. $e^{i\pi} = \cos\pi + i\sin\pi = -1 \Rightarrow e^{i\pi} + 1 = 0$

c. $e^{i\pi} = 1$

d. Euler's formula is $e^{ix} = \sin x + i\cos x$

40 [Challenging] • Target: Modeling with logarithms: the Richter scale • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$

41 [Challenging] • Target: Converting between exponential and logarithmic form • Cross-Lesson: 4.01 Inverse & Composite Functions

The function $f(x) = x - e^x$ has exactly one critical point. Find it.

a. $x = e$

b. $x = 1$

c. $x = -1$

d. $x = 0$

42 [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Convert to logarithmic form: $3^4 = 81$.

a. $\log_4 81 = 3$

b. $\log_3 81 = 4$

c. $\log_{81} 3 = 4$

d. $\log_3 4 = 81$

43 [Challenging] • Target: Visualizing logarithmic functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Express $\log_3 7$ using the change of base formula with natural log.

a. $\frac{\ln 7}{\ln 3}$

b. $7\ln 3$

c. $\frac{\ln 3}{\ln 7}$

d. $\ln 7 - \ln 3$

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44 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$

c. $g(x) = 10^x$

d. $g(x) = \log x$

45 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The function $f(x) = e^x - x - 1$ satisfies $f(x) \geq 0$. Prove it.

a. Only for $x \geq 0$

b. $f(0) = 0$, $f'(x) = e^x - 1 = 0$ at $x = 0$ (minimum). $f''(0) = e^0 = 1 > 0$: global min at $x = 0$ with value 0.

c. Not true

d. By AM-GM

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